Car Brand and Model Detection Using Deep Learning

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***Abstract*—Car brand and model detection problem is a difficult visual computer science problem in terms of similarity between cars' exterior design and changing environment. In this study, a deep learning-based system capable of automatically identifying car brand and model information from a single image is proposed. This project provides hierarchical two stage classification structure which detects car model after predicting car brand. Custom ResNet from scratch based deep learning is used as model architecture. Dataset that is used in project is collected from websites by web scraping techniques and with manually taken photos to reflect real world scenarios. Two staged hierarchical model structure has 81% validation accuracy in total and it is better than other experiments with single classifier model, less layer model, more learning rate model, and alternative Custom ResNet model that are built to make comparison and experiments.**

Keywords—deep learning, vehicle brand and model recognition, convolutional neurol networks (cnn), classification, residual neurol networks

I. INTRODUCTION

Computer vision and deep learning techniques have achieved significant success in object recognition and classification problems and these technologies are used actively nowadays. One of the areas of these technologies are car brand and model detection. Smart traffic systems, security applications, car park management, insurance processes and urban transport analysis can use brand and model detection. However, some problems such as visual similarity between cars, instability of environment conditions make car brand and model detection difficult in terms of research topic.

Cars have similar visual characteristics, especially when considering different models from the same manufacturer or different generations of the same model. Factors such as lightning conditions, background variety, camera angle and shadows make this classification problem more complex. Also, in some situations, logos of brands can be hidden, resolution of image can be low. So, only classifying cars by looking at the brand logos are not efficient. For these reasons, learning-based methods are needed that take into account both the general car structure and its distinctive details.

In this project, the main idea is building deep learning-based system that can detect car brand and model automatically from a single car image. Approach to build this system is hierarchical structure, which detects brand of the car before predicting car model. After brand detection, pipeline selects one of the car model classifiers to predict car model as well. With this idea, aim is to reducing class number and improving accuracy comparing to single car brand and model detection approach [1].

Dataset that is used in this project is collected with web scraping techniques. Aim is creating a dataset with images that have different camera angles, environment conditions, lightning, background etc. Real world conditions are reflected to get realistic experiment results.

As the model architecture, custom ResNet-based convolutional neural networks designed from scratch were used. Power of skip connections is tried to use to generate more stable classification models.

The main objective of this study is to examine the strengths and weaknesses of a deep learning-based hierarchical approach to car brand and model detection under realistic conditions. The aim is for the findings to contribute to improving the reliability and usability of AI-based systems in real-life car recognition scenarios.

II. RELATED WORKS

A. Vehicle Brand Detection Using Faster R-CNN

A deep neural network-based classification algorithm has been presented by Kunduracı and Kahramanli [2] as a way to identify automobile brands. They discovered that the deep neural network-based Faster R-CNN approach achieves 67.66% accuracy.

Two submodules make up Faster R-CNN. A deep fully convolutional network in the first module suggests regions, which are then used in the second module. The Faster R-CNN component is the second module. The first module, known as RPN, obtains the feature network. RoI pooling is used to downsize proposed regions. The work they have done merely identifies the car's brands, not its model.

B. CarNet.AI

LastStar Ltd. created CarNet.AI [3], an enterprise-level API that can identify car models and brands. Its internal work cannot be located because it is not an open-source project, although they have included some information on their description page.

They initially identify each car in the picture and around them with a bounding box. The next stage is to use the bounding box dimensions to filter out the bounding boxes that are unsuitable for detection. Choosing a detection method is the third stage. There are three different approaches. They can identify every bounding box, the one nearest to the image's center, or only the largest bounding box. Finding the subclasses is the fourth step. Each bounding box chosen in the third phase is processed, and their classes are determined. Bounding boxes are only forwarded to the final stage if they have the "vehicle" class. They locate the vehicles' characteristics in the last step. These characteristics include the car's model, brand, color, and angle.

C. Vehicle Make and Model Recognition Using Local Features and Logo Detection

A two-stage approach for identifying car models and brands was presented by Tafazzoli and Frigiue [4]. To achieve high categorization accuracy, they employed Multiple Instance Learning. In order to increase the accuracy of the first stage, they only identify the brand from the car's logo in the second stage. In the first stage, they identify both the brand and the model of the vehicle. A supervised learning approach called Multiple Instance Learning addresses ambiguously labeled data in regression and classification problems. Unlike conventional supervised learning, which uses fixed-length feature vectors as instances, Multiple Instance Learning examines packets of instances within each packet.

Name	Brand Detection	Model Detection
Kunduracı et Al.	**Yes**	No
CarNet.AI	**Yes**	**Yes**
Tafazzoli et Al.	**Yes**	**Yes**
Our Work	**Yes**	**Yes**

Table-1: Related Works

III. DATASET PREPARATION

Dataset that is used in this project is prepared to learn fine-grained visual discrimination required for car brand and model detection problem. Web scraping techniques are used to collect real car images from an open-source web page [5]. Photos containing personal data, those of poor quality, and those where the car image was not visible were manually removed. Also, 10 photos are manually taken and added to dataset (5 images from Renault Kangoo (Figure-1), 5 images from Volkswagen Passat). With this approach, images in dataset have different environment conditions, lightnings, backgrounds, camera angles and image quality to simulate real world scenarios.

Dataset includes 8 different type of cars and 4 car brands. Each car brand has 2 car models. Car brands and models are Ford F-150 (thirteenth generation), Ford Mustang VI (Figure-2), Renault Kangoo I, Renault Twizy Z.E., Toyota Land Cruiser J40, Toyota Supra A80, Volkswagen Passat B6 and Volkswagen Type 1.

Figure-1: Renault Kangoo I, Manually Taken Photo

Figure-2: Ford Mustang VI from Web

	class_idx	class_name	train_n	val_n	train_percentage	val_percentage
0	3	Renault_Twizy_Z.E	196	49	16.88	16.84
1	0	Ford_F-150_(thirteenth_generation)	188	47	16.19	16.15
2	1	Ford_Mustang_VI	155	39	13.35	13.40
3	7	Volkswagen_Type_1	155	39	13.35	13.40
4	4	Toyota_Land_Cruiser_(J40)	149	37	12.83	12.71
5	5	Toyota_Supra_(A80)	126	31	10.85	10.65
6	2	Renault_Kangoo_I	106	27	9.13	9.28
7	6	Volkswagen_Passat_B6	86	22	7.41	7.56

Table-2: Dataset Class Names and Train/Validation Size

Brand detection classification model dataset is split by 80% and 20% in terms of training and validation (Table-2).

Train size: 1161 images

Test size: 291 images

Most of the websites have security configurations for scripts and bots to prevent unusual requests. This is the reason why image sizes of classes are not equal. Also, this situation reflects real world scenarios because a situation where all the class sizes are equal not possible most of the time.

The model-level dataset for vehicle brand classification was regrouped to obtain four main brand classes. This structure constitutes the first stage of the hierarchical brand-model approach. The distribution obtained for the brand classification dataset is as follows (Table-3):

Ford: 343 train, 86 validation images

Renault: 302 train, 76 validation images

Toyota: 275 train, 68 validation images

Volkswagen: 241 train, 61 validation images

	brand_idx	brand_name	train_n	val_n	train_percentage	val_percentage
0	0	Ford	343	86	29.54	29.55
1	1	Renault	302	76	26.01	26.12
2	2	Toyota	275	68	23.69	23.37
3	3	Volkswagen	241	61	20.76	20.96

Table-3: Brand Detection Classifier Dataset Train/Validation

Same dataset is used to train 4 car model prediction classifiers which is run after brand prediction classifier. This structure ensures a balanced and consistent learning environment for each brand during the model classification phase. Because brand information was known before, car model classifiers were able to learn more subtle visual differences by working with a smaller number of classes.

In conclusion, the dataset used in this project provides a solid foundation for investigating the car brand and model detection problem with its controlled class distribution, realistic diversity, and hierarchical classification structure.

IV. METHODOLOGY

Firstly, all images are resized to a standard input size which is 224x224 and they are converted to PyTorch tensors. Then, dataset is loaded via ImageFolder and class labels are assigned to the images. Data augmentation techniques which include random resized cropping, horizontal flipping, color jittering, slight rotation are applied to this dataset to increase variation and improve generalization.

Our approach adopts a hierarchical classification strategy which means that 4 car brands are classified firstly and outputs of this classification are classified as brands of these models accordingly instead of using a model which classify 8 models at once. Consequently, five models were developed (one for brand classification, four for model classification.).

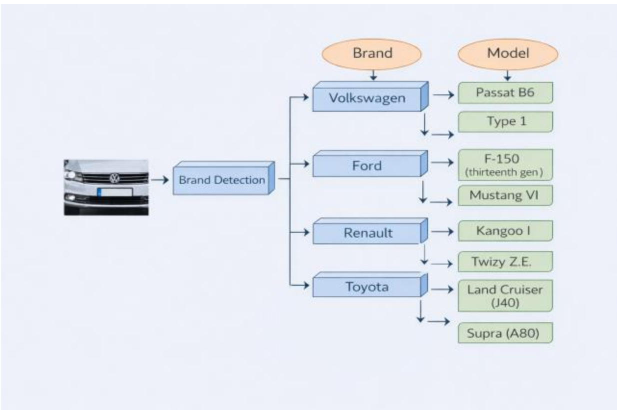

Figure-3. Hierarchial Classification Approach

All classifiers are inspired by ResNet-18 but are custom-designed. The architecture of our Custom ResNet Model is as follows:

- Firstly, a 7x7 convolution layer is used. Then batch normalization, ReLU and 3x3 max pool are applied.
- Four residual stages with two residual blocks which have 64, 128, 256, 512 channels in order are used.
- An average pooling layer is applied.
- To reduce overfitting, dropout (p = 0.3) is applied.
- A layer that gives output of class scores.

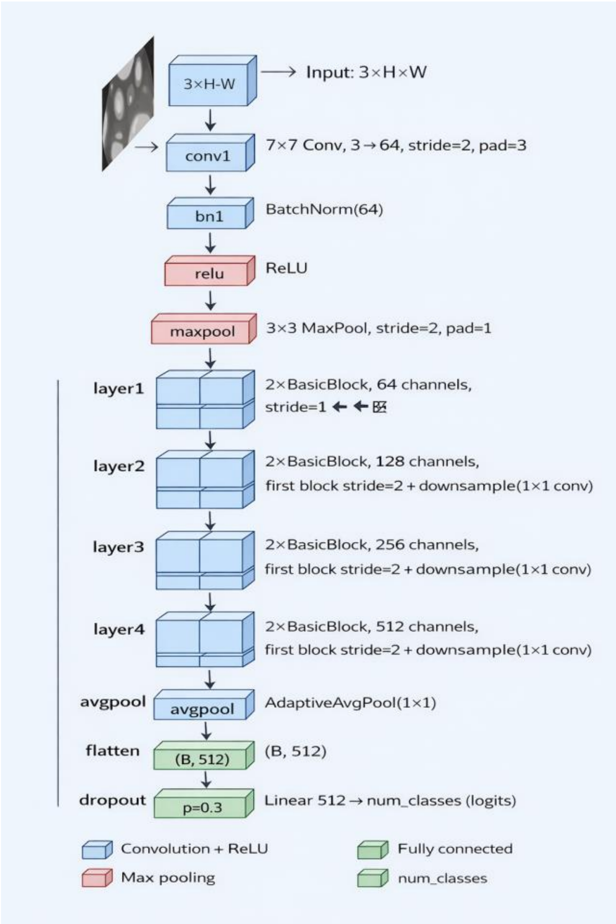

Figure-4. Architecture of Custom ResNet Model

Training is handled in the train_model function in the code. AdamW is used as the optimizer and Cosine Annealing is used to reduce the learning rate gradually. Learning rate is 3e-4 (3e-2 in Alternative Custom ResNet). We trained the brand ResNet model with 30 epochs and the model ResNet model with 20 epochs.

We also developed two alternative Custom ResNet models, one has one less residual stages, and the other one has one less residual stages and a higher learning rate (named as Alternate Custom ResNet Model). The aim of developing these two models is to observe the importance of the learning rate and the number of layers.

V. EXPERIMENTS AND RESULTS

Our first experiment is about the whether our hierarchical approach gives a better result than the approach the classifies all models at once. The validation accuracy of our brand classifier model is 0.83. The validation accuracies of our model classifiers are 0.99, 0.99, 0.99 and 0.97 respectively. The results of the model classifiers are close to 1, so it can be said that classifying models after brands is a good approach which is our main claim.

We also calculated the total accuracy of our model to compare with single classifier model. To find total accuracy, all validation accuracy scores for each brand are multiplied with brand classifier validation accuracy and they are also multiplied with the participation percentage of each brand to the dataset. The formula in the below represents this calculation:

$$TotalAccuracy = \sum_{i=1}^{4} p_i \times Acc_{brand} \times Acc_{model_i}$$

Total Accuracy is calculated as $0.29 \times 0.83 \times 0.99 + 0.26 \times 0.83 \times 0.99 + 0.23 \times 0.83 \times 0.99 + 0.22 \times 0.83 \times 0.97 \approx 0.81$. It is slightly better than the single classifier model which has 0.79 accuracy. It is expected that the gap between these accuracies would be higher if the experiment were conducted with more car brands and models. The brand classifier accuracy scores according to epochs for our Custom Resnet and single classifier can be found below:

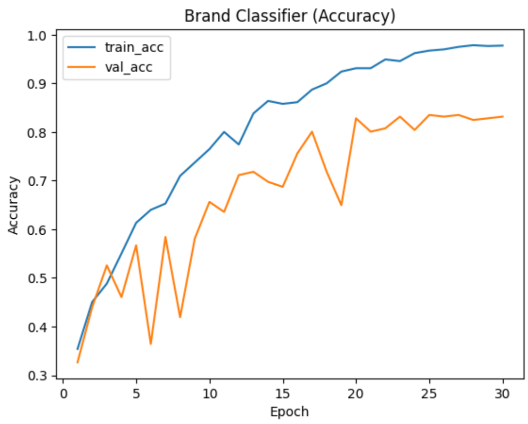

Figure-5. Accuracy Plot of Custom ResNet Brand Classifier

Figure-6. Accuracy Plot of Single Classifier Brand Classifier

Our second experiment is about comparing our Custom ResNet model with Custom ResNet with Less Layers and Alternative Custom ResNet model. The brand classifier accuracy is 0.76 for Custom ResNet with Less Layers and 0.59 for Alternative Custom ResNet model which are worse than our Custom ResNet model as expected. Accuracy scores for these two models can be found below:

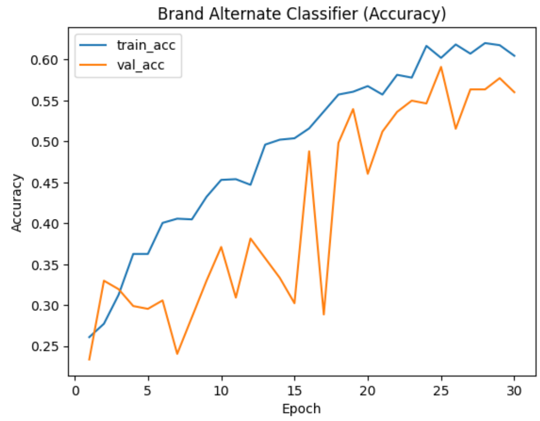

Figure-7. Accuracy Plot of Alternative Custom ResNet Brand Classifier

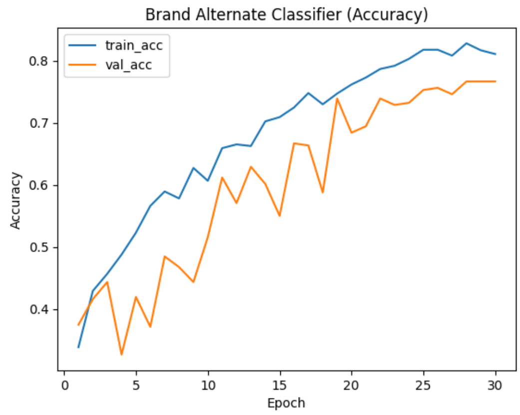

Figure-8. Accuracy Plot of Custom ResNet with Less Layers Brand Classifier

It can be said that models with less layers are not efficient as much as our main custom ResNet model. We wanted to analyze and see importance of the layer numbers and learning rate in same experiment.

Also, a confusion matrix is plotted to visualize which brands are predicted correctly and which brand is predicted for the incorrect predictions. Lastly, PCA is used to project learned feature representations to a lower latent space for analysis. All these visuals are plotted for the car model classifiers as well. The plots are given below:

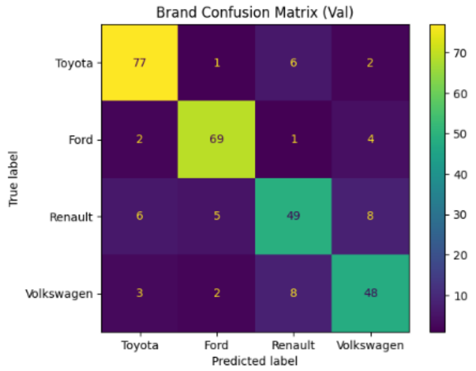

Figure-9. Confusion Matrix of Custom ResNet Brand Classifier

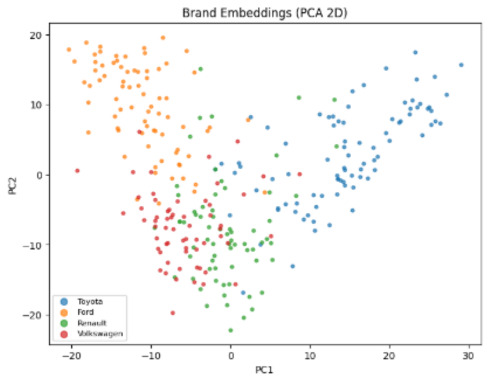

Figure-10. PCA plot of Custom ResNet Brand Classifier

All these viusals are plotted for the car model classifiers as well and can be found on the uploaded codes [1].

VI. CONCLUSION

In this project, the car brand and model detection problem is addressed using a deep learning-based approach, and the effectiveness of a hierarchical classification structure is investigated through detailed experiments. The goal is to accurately distinguish between different car brands and models under different environmental conditions. For this purpose, brand and model detection tasks are solved in a two-stage pipeline. In the first stage, the car brand is predicted, and in the second stage, the relevant car model is classified based on the predicted brand.

Experimental results have shown that the hierarchical approach offers higher overall accuracy compared to the single-stage structure (classifying all models simultaneously). The hierarchical structure yielded 81% overall validation accuracy, while identifying all car make and model in a single model yielded 79% validation accuracy. It is expected that expanding the dataset will yield even better results than classifying in a single model, as the hierarchical structure reduces the number of classifications.

The study also compared different Custom ResNet architectures. It was observed that alternative models with fewer layers or higher learning rates performed worse than the our main Custom ResNet model. This shows that network depth and appropriate hyperparameter selection are critical in fine-grained classification problems such as car brand and model detection. Experiments revealed a direct correlation between architectural depth and learning stability.

In addition, the learning behavior of the models was analyzed using confusion matrices and PCA visualizations. These visualizations show that the distinction between brands and models is largely successful, while erroneous predictions generally occur among the classes that are most visually similar.

In conclusion, using an unique dataset and a custom ResNet-based architectures, hierarchical classification strategy provides an efficient solution to the car brand and model recognition challenge. According to the results, studies with bigger datasets that include more brands and models will show the benefits of this strategy even more.

REFERENCES

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